

Processes Communicating

Process: program running within a host

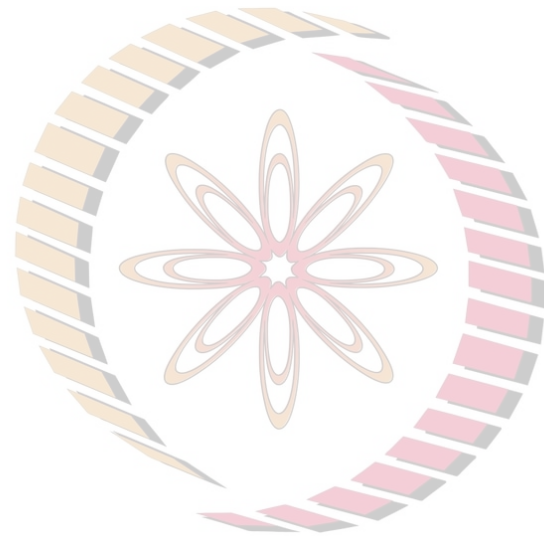
- ❑ Within same host, two processes communicate using **inter-process communication** (defined by OS)
- ❑ Processes in different hosts communicate by exchanging **messages**

clients, servers

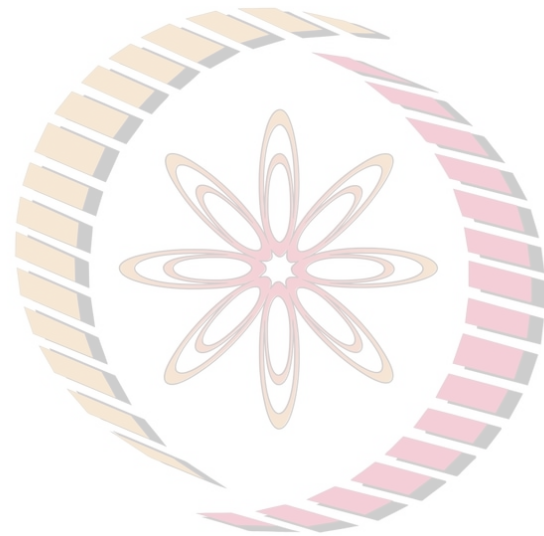
client process: process that initiates communication

server process: process that waits to be contacted

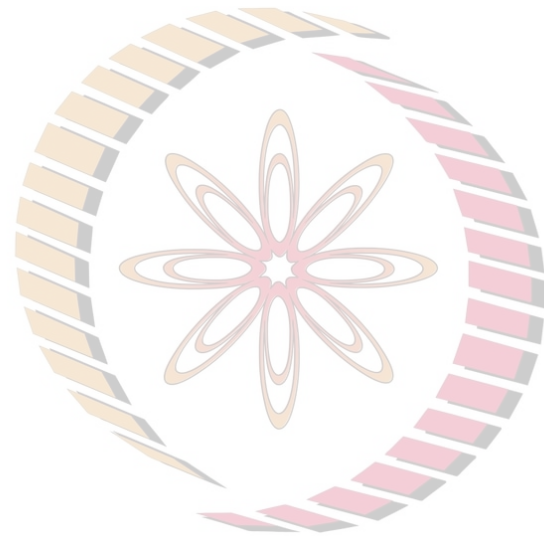
- ❑ note: applications with P2P architectures have client processes & server processes



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Application Protocols Types

- ❑ Text based protocols: exchange data in a format that humans can read using a simple text editor.
 - ❑ Character encoding (typically ASCII or UTF-8)
 - ❑ HTTP/1.1, DHCP
- ❑ Binary protocols: exchange data as a stream of bytes (1s and 0s) that follow a strict, pre-defined structure. They are not intended to be read by humans without a decoder.
 - ❑ Data is packed into fixed-length fields or "frames." For example, the first 4 bits might represent the version, the next 8 bits the length, and so on.
 - ❑ Ex: HTTP/2, HTTP/3, DNS, MQTT

Text based Protocol

Client request:

GET /hello.txt HTTP/1.1

User-Agent: curl/7.16.3 libcurl/7.16.3 OpenSSL/0.9.7l zlib/1.2.3

Host: www.example.com

Accept-Language: en, mi

Server response:

HTTP/1.1 200 OK

Date: Mon, 27 Jul 2009 12:28:53 GMT

Server: Apache

Last-Modified: Wed, 22 Jul 2009 19:15:56 GMT

ETag: "34aa387-d-1568eb00"

Accept-Ranges: bytes

Content-Length: 51

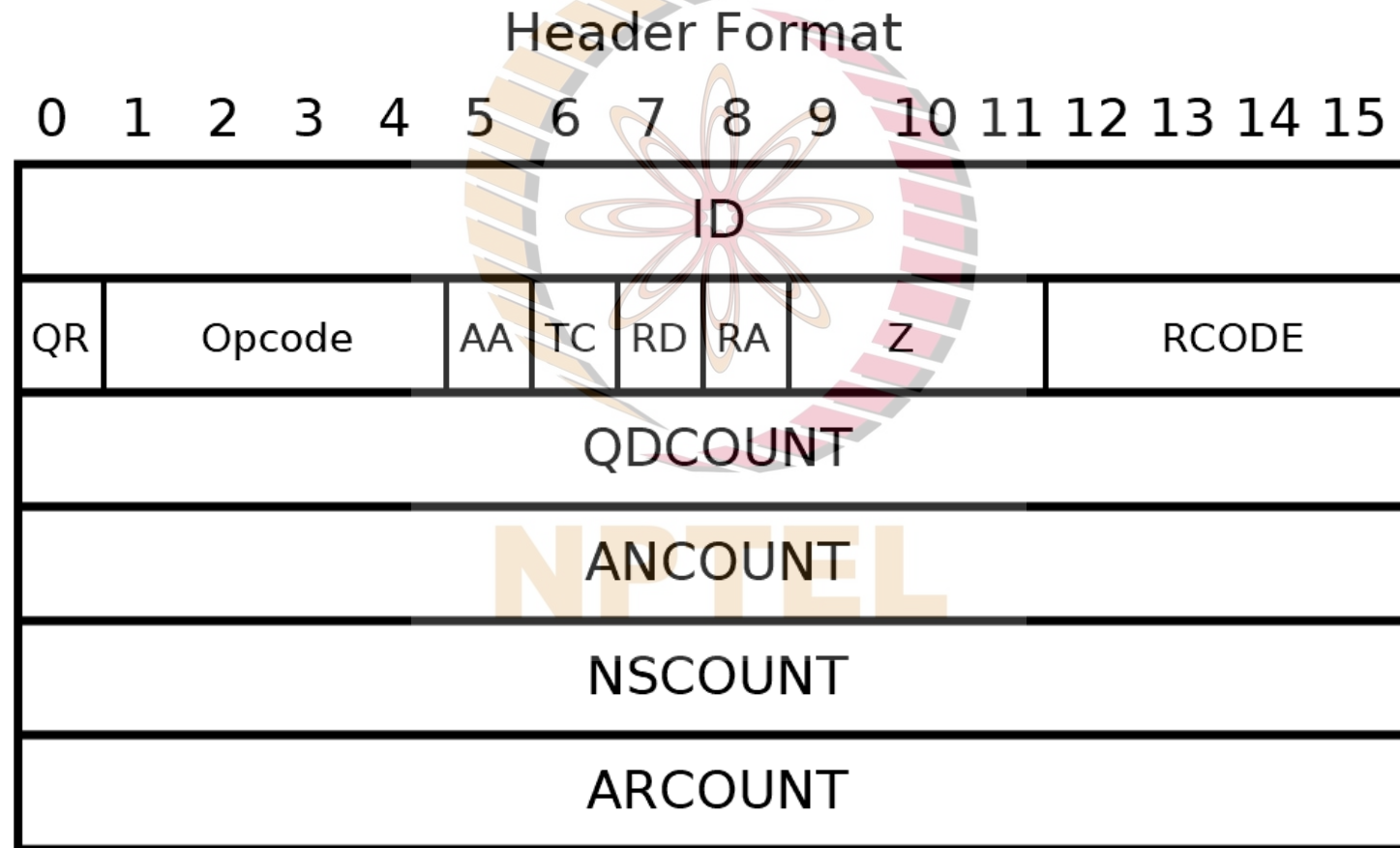
Vary: Accept-Encoding

Content-Type: text/plain

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Binary Protocol



Application Protocols Types: Text based Protocols

- ❑ **Transparency:** You can "sniff" the network traffic and understand exactly what is happening without special tools.
- ❑ **Ease of Debugging:** Developers can manually type commands into a terminal (like Telnet) to test a server.
- ❑ **Extensibility:** Adding new headers or commands is easy because you just add new words.

Application Protocols Types: Binary Protocols

- ❑ **Efficiency:** They use significantly less bandwidth because numbers and states are represented by bits rather than long text strings.
- ❑ **Speed:** Machines can parse binary data much faster than they can "read" and interpret text.
- ❑ **Robustness:** Because the structure is rigid, there is less room for the "interpretation errors" that sometimes plague text parsing.

Shift to Binary

- ❑ **Evolution for Scale:** While the early internet prioritized human-readability for research and debugging, modern architecture focuses on handling billions of concurrent devices.
- ❑ **Reduced Protocol Overhead:** Text-based headers (e.g., Content-Type: application/json) are repetitive and bulky; binary protocols replace these with compact, indexed tokens.
- ❑ **Performance Optimization:** Binary framing allows for header compression (HPACK) and multiplexing, significantly reducing the amount of data transmitted over the wire.
- ❑ **Lower Latency:** By squeezing maximum efficiency out of every packet, binary protocols like HTTP/2 and HTTP/3 minimize the time it takes for a user to receive data.
- ❑ **Machine-First Efficiency:** Binary data is parsed directly by hardware and software without the intensive string-processing required for text, lowering CPU usage on servers and mobile devices.
- ❑ **Enhanced Robustness:** Strict binary structures eliminate the ambiguity and "interpretation errors" common in text-based parsing, leading to more secure and predictable network behavior.